

Constellium and TARMAC Aerosave Accelerate Circularity in Aviation

Constellium and TARMAC Aerosave, a pioneer in aircraft and engine storage, maintenance, and eco-responsible dismantling, recently completed a project aimed at exploring a full-circular model for aluminum aircraft components. The two companies successfully recycled and remelted aluminum from end-of-life (EOL) aircraft (Figure 1) into a new, high-performance material suitable for future aerospace applications. This project demonstrates that even complex aerospace aluminum alloys from end-of-life aircraft can be fully recycled into material for new aircraft applications, providing a compelling path forward for future innovation and sustainability and highlighting the potential of aluminum to support circular economy goals in aerospace, without compromising material performance.



Figure 1. Dismantling of an aircraft using a specialized cutting gantry. (Source: TARMAC Aerosave.)

The Challenge of Recycling

Aircraft manufacturing has long focused on utilization of lightweight materials to improve the performance and fuel economy of commercial planes. For example, as a manufacturer of innovative, value-added aluminum products, Constellium contributes to sustainable aviation through lightweight aluminum solutions that improve fuel efficiency and support the industry's transition to Sustainable Aviation Fuels (SAF). "By reducing aircraft weight, our materials help to lower CO₂ emissions while maintaining the highest performance and safety standard," said Frank Eberl, head of Customer Applications Engineering, Europe, Constellium. "We are also preparing for next-generation propulsion technologies, leveraging decades of experience in space-grade materials and innovation. In parallel, our closed-loop recycling systems recover and reintegrate production scrap directly into manufacturing, ensuring no performance loss and minimal waste throughout the aluminum life cycle."

As the aircraft sector intensifies efforts to reduce its carbon footprint, the recyclability of the materials it uses become increasingly important, especially since aluminum recycling only uses 5% of the energy required for primary production and emits 95% fewer CO₂ emissions. Capturing materials from decommissioned aircraft at the end of their use-life is especially vital in this regard. However, it comes with some inherent challenges.

"The main technical challenge lies in the complexity of aerospace alloys and the mix of materials used in aircraft," explained Eberl. "Aluminum components often come from various alloys that must be disconnected and carefully sorted to maintain their performance properties during remelting. Paints, sealants, and fasteners can also introduce impurities that complicate recycling."

Eberl noted that in order to increase the recycling rates for EOL aircraft, the industry needs to achieve the following:

- Improve alloy traceability and segregation during dismantling.
- Develop dedicated recycling processes capable of remelting mixed aerospace alloys without loss of quality.

- Strengthen collaboration across the value chain, from OEMs to recyclers, to design aircraft with recyclability in mind and create circular supply loops, as demonstrated by the Constellium and TARMAC Aerosave partnership.

One Stop for Aircraft Care and Recovery

TARMAC Aerosave is a leading aircraft and engine maintenance, repair, and operations (MRO) specialist, founded by Airbus, Safran Aircraft Engines, and Suez. Operating Europe's largest aircraft storage and maintenance facilities across three sites in Tarbes and Toulouse, France, and Teruel, Spain, the company specializes in aircraft care: aircraft storage before return to service and aircraft maintenance and transition, which account for 85% of the business.

"Our main activity is to preserve aircraft," explained Sébastien Medan, Infrastructure and HSE director of TARMAC Aerosave. "We take care of the aircraft during their storage thanks to our MRO agreements and our large parking capacities and then return them to the skies when the owners request them back. We really are a one-stop shop expert in aircraft care through the entire life cycle."

Aircraft and engine recycling was actually TARMAC's first activity, leading to the development of unique processes. As the world leader in aircraft recycling, the company continues to innovate their process to improve its already exceptional recovery rate of 92%. To date, TARMAC Aerosave has received over 1,850 aircraft, redelivered 1,400 of them, and dismantled 450 aircraft and 255 engines.

"The circularity of aeronautical materials is at the heart of TARMAC Aerosave's DNA," said Alexandre Brun, president and CEO of TARMAC Aerosave. "It's what built its worldwide reputation as a leader in aircraft recycling, before becoming a one-stop shop. Aluminum is one of the materials that we have always sought to sort and recover in the best possible way in order to increase our recycling rate... We're proud to have joined forces with partners who share our commitment to advancing circular solutions for aviation."

Collaborative Research

An earlier aircraft recycling initiative, the PAMELA project, was launched in 2005 by Airbus, Suez, TARMAC Aerosave, and other partners. The initiative was the first large-scale European initiative to take into account the entire life cycle of aircraft dismantling and recycling. The PAMELA project showed that 85% of an aircraft's materials can be recovered and reused, laying the foundation for today's circular practices. The industrial realization of this project was the formation of TARMAC Aerosave, which has since increased the recyclability rate to over 92%.

Building on that groundwork, Constellium and TARMAC Aerosave launched their new collaboration in 2023, building on a shared ambition to advance circularity in aerospace. The new project combined TARMAC Aerosave's expertise in dismantling and material recovery with Constellium's capabilities in aluminum recycling, metallurgy, and alloy development.

The project ran for 18 months, with the support of Airbus. The two companies initiated the project with a review of the alloy types that would be explored, developing detailed knowledge of the chemical composition and potential contaminants, such as coatings or fasteners. This analysis ensured that the recycled aluminum would meet the required aerospace standards.

Once this was completed, TARMAC Aerosave dismantled and segregated materials from the retired commercial aircraft. According to Sabine Couhet-Guichot, Infrastructure & HSE deputy, TARMAC Aerosave, the dismantling and recovery process occurs in four process steps. "When the aircraft arrives, we drain its fluids and assign it a place in one of our recycling areas," she explains. "Once the aircraft has changed administrative status from 'aircraft' to 'waste' and has officially become our property, we begin by removing all high-value parts that the owner wishes to keep (engines, landing gear, flight instruments, etc.). Next, comes the stage we call 'greening,' which involves restoring the cabin to the green color it had when it was built, without any trim or equipment. We then dismantle and sort all the components: seats, carpets, walls, luggage compartments, cables, screws, etc. The final stage is to cut up the fuselage using a special gantry of our own design, which allows the aircraft to be sprayed with cold water throughout the operation to prevent dust and fire hazards. The sorted materials are then sent for recycling. In total, 92% of the aircraft's mass is recovered."

The aluminum recovered from the aircraft was pre-processed and shredded by MTB Recycling, decoated by ValoER (Figure 2) sorted by STEINERT, and finally delivered to Constellium, where the material was prepared for



Figure 2. Shredded and decoated aluminum recovered from the retired aircraft. (Source: Constellium.)

remelting and alloy production. "These shredding and sorting steps were key differentiators of the collaboration compared to the earlier PAMELA project, where entire aircraft sections were directly remelted without alloy separation," noted Guillaume Bes, Metal & Recycling Program manager, Constellium. Airbus supported the project, validating its potential for aerospace reuse.

The project reached its first milestone in mid-2025, when Constellium successfully remelted the aircraft scrap and cast it into four to five aluminum ingots at lab scale at its C-TEC research and technology center (Figure 3). "We recycled 2xxx aluminum alloys from EOL fuselages, because they are the most commonly used alloys in the current fuselage set to be dismantled," said Bes. "Because of their complex composition, 2xxx alloys are challenging to recycle without quality loss and are often downcycled into lower-performance applications. Through optimized shredding, advanced sorting, decoating, and precise remelting, the teams were able to maintain alloy integrity, demonstrating what's possible through collaboration and innovation across the value chain."



Figure 3. An aluminum ingot produced from EOL aircraft scrap. (Source: Constellium.)

At the C-TEC research center, the recycled metal underwent a full series of chemical, mechanical, and metallurgical tests, including chemical composition analysis to verify alloy purity and control trace elements—which are a key step to achieving full industrial scale of EOL aircraft recycling. The recycled aluminum met aerospace-grade specifications, demonstrating that even complex alloys like the 2xxx series can retain their high-performance properties after recycling. These results confirm the technical viability of full circularity for aluminum in aerospace applications.

Next Steps

Following months of collaborative research and development, Constellium and TARMAC Aerosave were able to recover and remelt aluminum from retired commercial aircraft. "Our collaboration demonstrates that aluminum recovered from EOL aircraft can be fully recycled into new, high-performance aluminum, meeting aerospace-grade chemical and mechanical specifications," said Eberl. "This milestone brings the industry significantly closer to true circularity for aluminum in aviation."

The next step for the project will be to scale it up for industrial application and extend it to any aluminum alloy used in metallic aircraft, further demonstrating the full recyclability of end-of-life metallic airframes. "Repeatability of the process has now been validated at laboratory scale," shared Medan. "The next major step will be to carry out casting on the production line. TARMAC Aerosave is currently producing sufficient volumes of raw material to carry out the first industrial casting made with 100% of raw material coming from end-of-life aircraft." ■